

Exp No 1

Study/Demo of 4 axis robotic arm

Aim: Study/Demo of 4 axis robotic arm

A 4-axis robot is a machine with four joints that can move in multiple directions. It can be used to automate complex tasks with precision.

How does a 4-axis robot work?

- A 4-axis robot has four motors, with one in the wrist.
- The motors allow the robot to move along the x, y, and z axes.
- The fourth motor in the wrist allows for rotation while moving along the x, y, and z axes.

Why use a 4-axis robot?

- A 4-axis robot is a good choice for automated equipment handling complex tasks with precision.

What is a robotic axis?

- A robotic axis is a degree of freedom or independent motion.
- Each link connection in an industrial robot is a robotic axis.
- Robotic axes are also known as robot joints.
- Each axis has a motor that powers the relative motion of the joint.

Other types of robots: Articulated robots, SCARA robots, Delta robots, and Cartesian robots.

Result:

